EXPANDING THE APERTURE: A CONVERSATIONAL BRIEFING AGENT THAT LEARNS WHAT YOU CARE ABOUT WITHOUT NARROWING WHAT YOU SEE

Kate Wang
Stanford University
katewang@stanford.edu

ABSTRACT

AI news moves faster than most people can keep up with, and the signal is scattered across papers, lab blogs, forums and business press. I propose a voice-first briefing agent that acts like a chief of staff. It delivers a short morning brief, asks what the user wants to dive into, holds a grounded back-and-forth on those items, and writes what it learned about the user's interests and preferred delivery style into memory. Each new brief must balance three objectives: stories that matter in the world, stories relevant to this user, and new topics that widen the user's view instead of reinforcing a filter bubble. To keep conversation latency low, the agent pre-fetches sources for likely deep dives before the brief. We will evaluate on a frozen AI-news corpus with hindsight importance labels, a panel of simulated users, and a multi-week self-study.

1 PROBLEM DEFINITION AND MOTIVATION

The problem. It's hard to keep up with what is happening in the world, especially in AI. I feel constantly behind the ball. The bottleneck is not access to information but the time and attention to filter it. Typing and scrolling are poor fits for the moments I do have, like my morning commute. What I want is an interactive back-and-forth with a chief of staff who briefs me by voice: here's what's happening, what do you want to dig into, and next time, remember what I cared about without over-indexing it.

Who it matters to. Anyone who must stay current in a fast-moving field with little time: students, researchers, operators and investors in AI. The same need applies to any domain with a daily firehose of news.

Why an agent, not a simpler system. A static daily summary or a ranked feed does not solve this. The task needs four agentic capabilities that interact:

- **Retrieval and ranking under competing objectives:** each brief trades off global importance, personal relevance and exploration of new topics.
- **Memory written from conversation:** the agent infers interests from what the user says and does in dialogue (asks to go deeper, skips, asks follow-ups). It also learns *how* the user wants information packaged (length, depth, technicality). It must decay stale interests and keep room for new ones.
- **Tool use avoidance to reduce latency:** a voice conversation feels broken past roughly a second of delay, so the agent decides ahead of time which sources to pre-fetch for each item. It searches live only when the cache cannot answer.
- **Grounded dialogue:** every claim in the brief and deep dive must be supported by a cited source. A confident but wrong news briefing is a real harm.
- **Fit with CS329Z:** the project spans core course topics: retrieval, tool use and memory in building agents; the data agents need (a frozen news corpus with hindsight labels); and evaluation and safety (simulated users, faithfulness).

2 Comparison with Existing Work

News recommendation and conversational recommenders. Personalized news is a classic recommender problem: Li et al. (2010) framed it as a contextual bandit that balances known interests with exploring new articles, and MIND (Wu et al., 2020) is now a standard benchmark. These systems learn from clicks and optimize engagement but not from conversation. Conversational recommenders (Christakopoulou et al., 2016; Jannach et al., 2021) show that asking users questions speeds up preference learning, but target items like movies or products, not news, multi-day memory, or the cost of narrowing.

Diversity and the filter bubble. Diversity reranking goes back to MMR (Carbonell & Goldstein, 1998) and topic diversification (Ziegler et al., 2005). Nguyen et al. (2014) measured how recommenders narrow exposure over time, and Kaminskas & Bridge (2017) survey novelty, serendipity and coverage. We borrow these metrics for an agent whose interest model is written from dialogue.

Memory and simulated users. MemGPT (Packer et al., 2023), Mem0 (Chhikara et al., 2025) and LaMP (Salemi et al., 2024) evaluate whether an agent recalls user facts, not whether its memory improves what it chooses to show over time. LLM agents can simulate believable users (Park et al., 2023) and now evaluate recommenders (Zhang et al., 2024), conversational recommenders (Wang et al., 2023) and tool-using agents (Yao et al., 2024), but Zhu et al. (2024) show these simulators can be unreliable, which is why we should ground these in a real user (me + friends).

Attribution. Gao et al. (2023) and Liu et al. (2023) show that LLM answers with citations are often unsupported by the sources they cite. We should use citation faithfulness as a required safety metric.

The gap. Commercial products such as ChatGPT Pulse (OpenAI, 2025), Google's Daily Listen (Google, 2025) and Huxe (Huxe, 2025) show demand for personalized briefings but publish no evaluation of importance coverage, filter-bubble effects or faithfulness. To our knowledge, no prior system combines (1) interests learned from spoken conversation, (2) explicit exploration to widen the user's view, and (3) pre-fetching to meet voice latency, or evaluates such a system against hindsight importance, simulated and real users, and faithfulness together. Our contribution is the agent and evaluation structure.

3 Next Steps and Direction

System design. The agent runs a daily loop: (1) *pre-compute*: ingest items from a fixed source list spanning research, products, business and policy, rank them by importance, relevance and novelty while reserving slots for exploration, and pre-fetch deep-dive material for likely follow-ups; (2) *brief* by voice and ask what to dive into; (3) *deep dive* from the pre-fetched cache with citations, searching live only when needed; (4) *update memory* with explicit and implicit interest signals and delivery preferences, decaying interests that have gone quiet. Voice uses off-the-shelf speech components; all quantitative evaluation runs on text transcripts.

Evaluation. We evaluate *selection* (did the agent pick the right stories, given what is important and what it knows about the user?) and *packaging* (did it deliver them the way this user wants?). Data is a frozen corpus from a past window (e.g. August–September 2026), which allows hindsight labels and full reproducibility: the primary importance label is a Hacker News score above a calibrated cutoff, the secondary is coverage by multiple sources from our list within 14 days. Evaluation has three layers: (1) user-independent metrics on the corpus (importance recall, citation faithfulness, latency and pre-fetch efficiency); (2) 10–20 simulated users with hidden, drifting interest profiles, one calibrated to me (relevance, interest recovery, topic coverage, packaging adherence); and (3) me as the primary real user for 4–5 weeks, with daily ratings and blinded baseline days. Baselines are an importance-only brief, a static interest profile, no exploration, and no pre-fetching.

How we will know it works. The full agent should beat the static profile on relevance while staying close to importance-only on importance recall, keep topic coverage above the no-exploration baseline, reach at least 95% supported claims, and answer most follow-ups from the pre-fetch cache. Targets will be refined after the first baselines run.

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A AI-USE DISCLOSURE STATEMENT

Ideation. I used Claude as a brainstorming partner to explore a couple of projects that are top of mind for me (including a voice-driven personal memory agent, a task-routing capture agent and an AI-news briefing agent). I chose the project I thought would be most interesting for the structure of the course and fleshed out what I thought would be feasible to achieve within the quarter (by myself, since I am team of 1).

Writing. Claude produced a first draft of this proposal from my direction and phrasing. I reviewed, edited and take full responsibility for the final text. Claude also helped find related work, which I read summaries of and picked the most relevant ones. Each reference was checked against publisher metadata (Crossref DOIs), arXiv or the original web page.

Code and experiments. No code or experiments exist yet at the proposal stage.